

SET

Sign-Equivalent Transformation

Nonlinear Equilibrium Systems and Network Games

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Research Seminar

The representation of an equilibrium system

Many economic models reduce to nonlinear first-order conditions.

The difficulty

The original representation may obscure monotonicity, a potential function, or a useful uniqueness condition.

- Which transformations preserve the **entire equilibrium set**?
- Which representation makes the relevant structure easiest to verify?
- What can we learn about nonlinear network amplification?

SET, Introduction, pp. 2–7

Plan of the seminar

1. **Equivalence and geometry**

Variational inequalities, rectangular domains, and admissible transformations.

2. **Useful transformed systems**

P matrices, potential functions, and production networks.

3. **Nonlinear strategic complements**

CES peers, spectral thresholds, centrality, and key players.

4. **Strategic substitutes**

Convex costs, participation, and network rankings.

Nash equilibrium as a variational inequality

Player i chooses x_i in a closed convex interval K_i . Payoff $u_i(\cdot, x_{-i})$ is differentiable and concave.

$$K = \prod_{i=1}^n K_i, \quad F_i(x) = -\frac{\partial u_i(x)}{\partial x_i}.$$

Lemma 1

$$x^* \in \text{NE} \iff \langle x - x^*, F(x^*) \rangle \geq 0 \quad \text{for every } x \in K.$$

The operator F collects marginal incentives. Its derivative matrix is $J_F = DF$.

The VI formulation includes both interior and boundary equilibria.

SET, Definition 1 and Lemma 1, pp. 8–9

Boundary optimality on a rectangular domain

For $K = \prod_i [\ell_i, u_i]$, each coordinate can vary independently.

Location of x_i^*	VI condition	Marginal payoff
Lower bound ℓ_i	$F_i(x^*) \geq 0$	$\partial_i u_i(x^*) \leq 0$
Interior	$F_i(x^*) = 0$	$\partial_i u_i(x^*) = 0$
Upper bound u_i	$F_i(x^*) \leq 0$	$\partial_i u_i(x^*) \geq 0$

Example: nonnegative actions

For $K = \mathbb{R}_+^n$: $x^* \geq 0$, $F(x^*) \geq 0$ and $x_i^* F_i(x^*) = 0$ for every i .

SET, Theorem 1 and its proof, pp. 9–10 and 43

Sign equivalence

Definition 2

Two operators F and $\tilde{F}$ are sign equivalent on K if

$$\operatorname{sgn} F_i(x) = \operatorname{sgn} \tilde{F}_i(x) \quad \text{for every } i \text{ and every } x \in K.$$

The transformation preserves

- Positive, zero and negative values.
- Each coordinate's incentive direction.

The transformation may change

- Magnitudes and derivatives.
- Monotonicity and integrability.

The equality holds on the **whole domain**, including away from equilibrium.

SET, Definition 2, p. 9

Theorem 1: equilibrium-set preservation

Suppose F and $\tilde{F}$ are sign equivalent on K .

Interior solutions on a general domain

$$\text{Sol}(K, F) \cap \text{int } K = \text{Sol}(K, \tilde{F}) \cap \text{int } K.$$

All solutions on a rectangular domain

If $K = \prod_i [\ell_i, u_i]$, allowing infinite endpoints, then

$$\text{Sol}(K, F) = \text{Sol}(K, \tilde{F}).$$

For interior solutions, even **zero equivalence** suffices: $F_i(x) = 0 \iff \tilde{F}_i(x) = 0$.

SET, Theorem 1 and Remark 1, pp. 9–10

Why the coordinate argument works

Fix a candidate $x^* \in K = \prod_i [\ell_i, u_i]$.

A one-coordinate deviation

For any feasible $t \in [\ell_i, u_i]$, choose $x = x^* + (t - x_i^*)e_i$. The VI gives

$$(t - x_i^*)F_i(x^*) \geq 0.$$

Its validity depends only on the sign of $F_i(x^*)$.

Sign equivalence therefore preserves every coordinate inequality.

$$\sum_i (x_i - x_i^*) \tilde{F}_i(x^*) \geq 0 \quad \text{for every } x \in K.$$

Reversing the argument proves equality of the solution sets.

SET, proof of Theorem 1, p. 43

A square and a disk

The operators $F_a(x) = (a(x_1 - 2), x_2 - 2)^\top$, $a > 0$, are sign equivalent. Each is the gradient of

$$\phi_a(x) = \frac{a}{2}(x_1 - 2)^2 + \frac{1}{2}(x_2 - 2)^2.$$

Square: $[-1, 1]^2$

The unique minimizer is always

$$x^* = (1, 1).$$

The solution is independent of a .

Disk: $x_1^2 + x_2^2 \leq 1$

$$x^*(1) = (1/\sqrt{2}, 1/\sqrt{2}),$$

$$\lim_{a \rightarrow \infty} x^*(a) = (1, 0).$$

The solution varies with a .

SET, Example 1 and footnote 18, pp. 10–11

Theorem 2: the domain restriction is substantive

Universal preservation

On full-dimensional, nonempty closed convex domains, the paper's characterization is

every sign-equivalent pair has the same VI solutions $\iff K$ is rectangular.

- Separate interval constraints fit the theorem directly.
- Coupled budgets and joint capacity constraints require additional analysis.
- A particular transformation may still work on a nonrectangular domain.

The necessity statement concerns **all transformations**, not one chosen pair.

SET, Definition 4 and Theorem 2, p. 11

SET1: positive componentwise rescaling

Proposition 1

$$\tilde{F}_i(x) = \alpha_i(x)F_i(x), \quad \alpha_i(x) > 0 \quad \text{for every } x \in K.$$

- The scale may differ across players and depend on the entire action profile.
- For $\tilde{u}_i = \psi_i(u_i)$ with $\psi'_i > 0$, $\tilde{F}_i = \psi'_i(u_i)F_i$.

The derivative structure can change

$$\nabla \tilde{F}_i = \alpha_i \nabla F_i + F_i \nabla \alpha_i.$$

Even a rescaling can remove or create a convenient matrix property.

SET, Proposition 1 and Remarks 2–3, pp. 12–13

SET2: monotone transformations of a difference

Write $F_i(x) = A_i(x) - B_i(x)$.

Increasing transformation h

$$\tilde{F}_i(x) = h(A_i(x)) - h(B_i(x)).$$

Decreasing transformation h

The order reverses to preserve the sign:

$$\tilde{F}_i(x) = h(B_i(x)) - h(A_i(x)).$$

For positive A_i, B_i , $h = \log$ turns a level comparison into a log comparison.

Compositions of SET1 and SET2 preserve sign equivalence.

SET, Proposition 1, p. 12 and Remark 4, p. 13

Uniqueness tools after transformation

Sufficient conditions for at most one solution

- $\tilde{F}$ is strictly monotone, or
- K is rectangular and $J_{\tilde{F}}(x)$ is a P-matrix for every $x \in K$.

P-matrix: all principal minors are strictly positive.

A convex potential

If $\nabla\phi = \tilde{F}$ and ϕ is convex, then

$$\text{Sol}(K, \tilde{F}) = \arg \min_{x \in K} \phi(x).$$

Strict convexity gives at most one minimizer.

Existence requires a separate argument, such as compactness or coercivity.

SET, Lemmas 2–3 and Remark 5, pp. 13–15

Payoff scaling leaves equilibrium unchanged

Consider two players with $x_i \in \mathbb{R}$ and $\delta > 0$:

$$u_i = x_i - \frac{1}{2}x_i^2 + \delta x_i x_j, \quad i \neq j.$$

Scale player 2's payoff by $\kappa > 0$. The two operators are

$$F(x) = \begin{pmatrix} x_1 - 1 - \delta x_2 \\ x_2 - 1 - \delta x_1 \end{pmatrix}, \quad \tilde{F}(x) = \begin{pmatrix} F_1(x) \\ \kappa F_2(x) \end{pmatrix}.$$

A scale-invariant benchmark

For every $\kappa > 0$ and $0 < \delta < 1$, SET recovers the unique equilibrium

$$x_1^* = x_2^* = \frac{1}{1 - \delta}.$$

SET, payoff-scaling example, pp. 15–16; κ replaces the example's β

Why a direct monotonicity test can be restrictive

The scaled derivative matrix and its symmetric part are

$$J_{\tilde{F}} = \begin{pmatrix} 1 & -\delta \\ -\kappa\delta & \kappa \end{pmatrix}, \quad S = \begin{pmatrix} 1 & -\frac{(1+\kappa)\delta}{2} \\ -\frac{(1+\kappa)\delta}{2} & \kappa \end{pmatrix}.$$

Since $S_{11} > 0$, positive definiteness requires

$$\det S = \kappa - \frac{(1+\kappa)^2\delta^2}{4} > 0 \quad \Longleftrightarrow \quad \delta < \frac{2\sqrt{\kappa}}{1+\kappa}.$$

Example: $\kappa = 4$

Direct monotonicity: $\delta < 0.8$. SET: $0 < \delta < 1$.

The P-matrix test also recovers $\delta < 1$ in this constant-scaling example.

SET, eqs. (7)–(8), pp. 15–16; symmetric-part calculation

Theorem 3: inversion and a network potential

For continuous, strictly monotone R_i , suppose

$$h_i(x_i) = R_i\left(s_i(x_i) + \delta \sum_j g_{ij}x_j\right).$$

Define $q_i = R_i^{-1} \circ h_i - s_i$ wherever the inverse exists.

$$\tilde{f}_i(x) = q_i(x_i) - \delta(Gx)_i = 0.$$

Symmetric G

$$\phi(x) = \sum_i \int_{x_i^0}^{x_i} q_i(t) dt - \frac{\delta}{2} x^\top Gx, \quad \nabla \phi = \tilde{f}.$$

Interior equilibria are stationary points of ϕ .

Decreasing R_i : zeros are preserved, but boundary VIs require a sign correction.

SET, Theorem 3 and discussion, pp. 23–26

Nonlinear best responses and spectral curvature

Let $x_i = v_i(\delta \sum_j g_{ij} x_j)$ with $v_i' > 0$. Wherever each inverse exists,

$$\tilde{F}_i(x) = v_i^{-1}(x_i) - \delta \sum_j g_{ij} x_j, \quad \phi(x) = \sum_i \int^{x_i} v_i^{-1}(t) dt - \frac{\delta}{2} x^\top G x.$$

The transformed Hessian is symmetric:

$$\nabla^2 \phi(x) = D(x) - \delta G, \quad D_{ii}(x) = \frac{1}{v_i'(v_i^{-1}(x_i))}.$$

A sufficient curvature condition

$$\inf_{i, x_i} D_{ii}(x_i) > \delta \lambda_{\max}(G) \implies \phi \text{ is strongly convex.}$$

This gives at most one equilibrium on the domain of a valid sign-equivalent VI.

SET, eqs. (25)–(30), Proposition 5 and Remarks 8–9, pp. 26–28

CES production networks

- $a_{ij} \geq 0$ measures input dependence, with $\sum_j a_{ij} = 1 - \alpha_i$.
- Labor share $\alpha_i > 0$, productivity $z_i > 0$, wage $w > 0$.
- Input substitution elasticity $\sigma_i \geq 1$.

Equilibrium prices

For $0 < \alpha_i < 1$ and $\sigma_i > 1$, cost minimization gives

$$p_i z_i = C_i(p), \quad C_i(p) = w^{\alpha_i} \left(\frac{\sum_j a_{ij} p_j^{1-\sigma_i}}{1 - \alpha_i} \right)^{\frac{1-\alpha_i}{1-\sigma_i}}.$$

The Cobb–Douglas case $\sigma_i = 1$ follows by a limit.

Question: Does this nonlinear price system have exactly one solution?

SET, Section 3.1, equations (9)–(13), pp. 18–19

Log SET and the row-sum bound

$$p_i z_i - C_i(p) \sim_{\text{sign}} \log(p_i z_i) - \log C_i(p), \quad x_i = \log(p_i/w).$$

$$\tilde{F}_i(x) = x_i + \log z_i - \frac{1 - \alpha_i}{1 - \sigma_i} \log\left(\frac{\sum_j a_{ij} e^{(1-\sigma_i)x_j}}{1 - \alpha_i}\right).$$

A uniform margin supplied by labor

$$D\tilde{F}(x) = I - W(x), \quad W_{ij}(x) = \frac{(1 - \alpha_i) a_{ij} e^{(1-\sigma_i)x_j}}{\sum_k a_{ik} e^{(1-\sigma_i)x_k}}.$$

$W(x) \geq 0$ and $\sum_j W_{ij}(x) = 1 - \alpha_i < 1$ at every x . Thus $I - W(x)$ is a P -matrix.

$T(x) = x - \tilde{F}(x)$ is a contraction with modulus $1 - \min_i \alpha_i < 1$. A unique equilibrium exists.

SET, Proposition 2 and equation (14), p. 19

Cobb–Douglas and the Leontief inverse

As $\sigma_i \rightarrow 1$, let $\varepsilon_i = \log z_i$ and $A = (a_{ij})$.

The nonlinear system becomes linear

$$x + \varepsilon - Ax = 0, \quad x^* = -(I - A)^{-1}\varepsilon.$$

$$L = (I - A)^{-1} = I + A + A^2 + \cdots, \quad \rho(A) \leq \max_i (1 - \alpha_i) < 1.$$

- $-\partial x_i / \partial \varepsilon_j = L_{ij}$ measures the price response to productivity in industry j .
- Powers of A collect direct and indirect input paths.
- The same resolvent $(I - \delta G)^{-1}$ appears in Bonacich centrality. Here the forcing term is $-\varepsilon$ and A contains input weights.

Other applications of SET

Model	Transformation	Conditional result
Nonlinear peers	Log SET and log actions	Uniqueness conditional on existence (Proposition 3).
Oligopoly pricing	Positive rescaling	At most one equilibrium under E1–E3.
Public goods with income	Inverse Engel curve	Potential. Uniqueness under normality and spectral conditions.
Multiplex networks	Transform marginal benefits	Potential. Strict curvature needs spectral restrictions.

All results require the corresponding model assumptions.

SET, Propositions 3, 4 and 6, Sections 3.2–3.5, pp. 20–31

CES peer effects on an unnormalized network

$$u_i(x) = a_i x_i - \frac{1}{2} x_i^2 + \delta x_i \left(\sum_j g_{ij} x_j^\beta \right)^{1/\beta}.$$

Best response

$$T_i(x) = a_i + \delta S_i(x), \quad S_i(x) = \left(\sum_j g_{ij} x_j^\beta \right)^{1/\beta}.$$

- $a_i > 0$, $\delta > 0$, $\beta > 0$, and $x_i \geq 0$.
- G is an undirected adjacency matrix, without self-loops.
- We use a **connected, nonempty** network for the Perron-limit results.

SET, Section 4.1, pp. 32–33, equations (42)–(44)

What the CES parameter changes

For positive neighbor actions,

$$\frac{\partial S_i}{\partial x_j} = g_{ij} x_j^{\beta-1} S_i(x)^{1-\beta}.$$

For two linked neighbors j and k ,

$$\frac{\partial S_i / \partial x_j}{\partial S_i / \partial x_k} = \left(\frac{x_j}{x_k} \right)^{\beta-1}.$$

Relative peer influence

$\beta > 1$ favors high-action peers at the margin. $0 < \beta < 1$ places relatively more marginal weight on low-action peers.

Changing β also changes the scale of this unnormalized aggregate.

SET, pp. 32–34; the derivative ratio is an immediate calculation

A log transformation of the equilibrium conditions

Every equilibrium satisfies $x_i \geq a_i > 0$. Let $F_i(x) = x_i - a_i - \delta S_i(x)$.

SET2, followed by a change of variables

$$\operatorname{sgn} F_i(x) = \operatorname{sgn}[\log x_i - \log(a_i + \delta S_i(x))].$$

Set $z_i = \log x_i$ and write

$$H_i(z) = z_i - \log[a_i + \delta S_i(e^z)].$$

The logarithmic difference preserves signs. The coordinate change maps $\mathbb{R}_{++}^n$ bijectively to $\mathbb{R}^n$.

SET, Theorem 4 and the method of Proposition 3, pp. 20–21, 33

The transformed Jacobian and uniqueness

Let $S_i = S_i(e^z)$ and define

$$W_{ij}(z) = \frac{\delta S_i}{a_i + \delta S_i} \cdot \frac{g_{ij} e^{\beta z_j}}{\sum_k g_{ik} e^{\beta z_k}}.$$

A substochastic interaction matrix

$$DH(z) = I - W(z), \quad W(z) \geq 0, \quad \sum_j W_{ij}(z) = \frac{\delta S_i}{a_i + \delta S_i} < 1.$$

- $I - W(z)$ is a nonsingular M matrix and hence a P matrix.
- There is **at most one** positive equilibrium for every $\delta > 0$.
- The row-sum bound is pointwise. It does not establish existence on $\mathbb{R}^n$.

SET, Theorem 4 uniqueness argument, p. 33; expanded derivative calculation

The spectral existence threshold

Theorem 4

For $\bar{\delta} = \lambda_{\max}(G)^{-1/\beta}$:

- $0 < \delta < \bar{\delta}$: a unique positive equilibrium exists.
- $\delta \uparrow \bar{\delta}$: all actions diverge, with convergent ratios.
- $\delta \geq \bar{\delta}$: no finite positive equilibrium exists.

On a d -regular network with $a_i = a$,

$$x_i^* = \frac{a}{1 - \delta d^{1/\beta}}, \quad \bar{\delta} = d^{-1/\beta}.$$

SET, Theorem 4 and Example 2, pp. 33–34; equality case in proof, pp. 48–49

Why the threshold is necessary

At a positive equilibrium, powers are taken componentwise:

$$(x - a)^\beta = \delta^\beta G x^\beta < x^\beta.$$

Let $e \gg 0$ satisfy $Ge = \lambda_{\max}(G)e$. Since G is symmetric,

$$\delta^\beta \lambda_{\max}(G) e^\top x^\beta < e^\top x^\beta.$$

A strict restriction

$$\delta^\beta \lambda_{\max}(G) < 1 \iff \delta < \bar{\delta}.$$

The positive baseline makes the inequality strict, excluding the critical point.

SET, Theorem 4, pp. 48–49; simplified Perron-vector proof

Existence through a Perron upper bound

For $\delta < \bar{\delta}$, choose $z_i = Me_i^{1/\beta}$. Then

$$T_i(z) = a_i + \delta \lambda_{\max}(G)^{1/\beta} z_i \leq z_i$$

whenever

$$M \geq \max_i \frac{a_i}{(1 - \delta \lambda_{\max}(G)^{1/\beta}) e_i^{1/\beta}}.$$

Monotone iteration

$$x^{(0)} = a, \quad x^{(k+1)} = T(x^{(k)}), \quad a \leq x^{(k)} \leq x^{(k+1)} \leq z.$$

The limit exists and solves $x = T(x)$ by continuity. SET supplies uniqueness.

SET, proof of Theorem 4, p. 48; equivalent iteration from the lower bound

Comparative statics below the threshold

Monotone equilibrium responses

If $a' \geq a$ or $\delta' \geq \delta$, and both parameter values admit equilibrium, the corresponding equilibrium action vector weakly increases.

- The best-response map increases in neighbors' actions, a , and δ .
- The old equilibrium is a lower solution for the new best-response map.
- A new Perron upper bound makes the increasing iteration converge.

$$u_i(x^*) = \frac{1}{2}(x_i^*)^2.$$

SET, Theorem 4(i) and the equilibrium-payoff observation, pp. 33–34

Bonacich centrality as the linear special case

When $\beta = 1$,

$$x = a + \delta Gx.$$

The network multiplier

$$(I - \delta G)x = a, \quad x^* = (I - \delta G)^{-1}a.$$

- $a = \mathbf{1}$: a standard Katz–Bonacich centrality vector.
- Heterogeneous a : baseline-weighted centrality.
- In the notation $(I - \delta G)x = b$, the vector b is this same baseline.

SET, discussion after Theorem 4, p. 34; Ballester et al. (2006)

Walks, invertibility, and amplification

If $\delta\rho(G) < 1$,

$$(I - \delta G)^{-1} = \sum_{k=0}^{\infty} \delta^k G^k, \quad x^* = a + \delta G a + \delta^2 G^2 a + \dots$$

- $(G^k)_{ij}$ counts length- k walks in an unweighted graph.
- Walks may revisit nodes and edges. Their weights are δ^k .
- The spectrum determines whether this infinite feedback sum converges.

Two distinct conditions

Invertibility requires $1 - \delta\lambda_k \neq 0$ for every k . The convergent nonnegative multiplier requires $\delta\rho(G) < 1$.

SET, p. 34; Neumann-series interpretation of the linear benchmark

The inverse acts separately on each eigenvector

Let $Gv_k = \lambda_k v_k$ with an orthonormal eigenbasis. Write

$$a = \sum_{k=1}^n (v_k^\top a) v_k.$$

Spectral decomposition

$$x^* = \sum_{k=1}^n \underbrace{\frac{1}{1 - \delta \lambda_k}}_{\text{amplification factor}} (v_k^\top a) v_k.$$

A baseline $a = cv_k$ stays in that eigenvector direction. Generic baselines combine several directions.

Standard spectral decomposition, connecting SET equation (47), p. 34

Eigenvector centrality at the critical limit

For the positive unit Perron vector e ,

$$Ge = \lambda_{\max}(G)e, \quad e \gg 0.$$

Theorem 4(ii)

$$\lim_{\delta \uparrow \bar{\delta}} \frac{x_i^*}{x_j^*} = \left(\frac{e_i}{e_j} \right)^{1/\beta}.$$

- $\beta = 1$: $x^* / \|x^*\|_2 \rightarrow e$.
- General $\beta > 0$: the vector $(x^*)^\beta$ approaches the Perron direction.
- Baseline heterogeneity affects levels, but disappears from these limiting ratios.

SET, Theorem 4(ii), pp. 33–34, equations (46)–(47); connected network

A three-node path: an exact nonlinear example



Set $a_i = 1$. Symmetry gives endpoint action L and center action C :

$$L = 1 + \delta C, \quad C = 1 + \delta 2^{1/\beta} L.$$

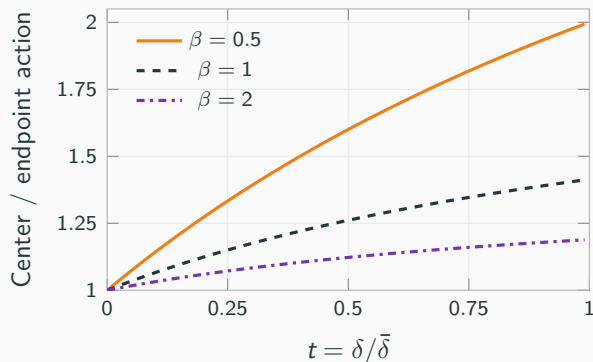
Closed-form solution

$$L = \frac{1 + \delta}{1 - \delta^2 2^{1/\beta}}, \quad C = \frac{1 + \delta 2^{1/\beta}}{1 - \delta^2 2^{1/\beta}}.$$

$$\lambda_{\max} = \sqrt{2}, \quad e \propto (1, \sqrt{2}, 1), \quad \text{and} \quad \bar{\delta} = 2^{-1/(2\beta)}.$$

An analytical example constructed for this seminar from SET's CES model

Relative actions along the path



$$\frac{C}{L} = \frac{1 + tq}{1 + t/q}, \quad q = 2^{1/(2\beta)}.$$

As $t \uparrow 1$

The ratio converges to q .

The action levels diverge.

At the same fraction of the threshold, smaller β produces a larger center advantage here.

Analytical curves for the seminar example; no finite equilibrium at $t = 1$

The key-player intervention

Let G^{-i} denote the network after removing player i .

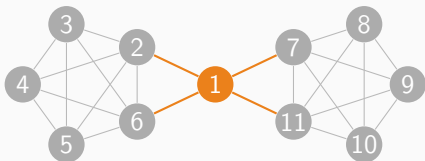
Total equilibrium activity lost

$$\Delta_i = \sum_j x_j^*(G) - \sum_{j \neq i} x_j^*(G^{-i}), \quad i^* \in \arg \max_i \Delta_i.$$

- Removing a player eliminates their action and changes everyone else's incentives.
- Highest equilibrium action need not identify the largest total effect.
- For general β , recompute equilibrium on each remaining network.

SET, key-player analysis, pp. 35–37; Ballester et al. (2006)

A bridge can matter more than the most active player



Linear benchmark, $\beta = 1$

δ	Key-player type
0.1	Node 2
0.2	Node 1

Nodes 2, 6, 7, and 11 have the highest degree and the highest action.

Nonlinear peers change the intervention ranking

At $\delta = 0.1$, the paper finds a switch from node 1 to the node-2 type near $\beta = 0.722$ within its reported parameter range.

SET, Figure 2 and key-player results, pp. 35–37; network redrawn from Figure 2

Network public goods with nonlinear costs

On an undirected network, player i chooses $x_i \geq 0$ and receives

$$u_i(x) = b\left(x_i + \delta \sum_j g_{ij}x_j\right) - c(x_i), \quad 0 < \delta \leq 1.$$

Assumption 2: benefits, costs, and a marginal crossing

$$\begin{aligned} b' > 0, \quad b'' < 0, \quad c'(x) > 0 \ (x > 0), \quad c'' \geq 0, \\ b'(0) > c'(0), \quad \lim_{x \rightarrow \infty} b'(x) < \lim_{x \rightarrow \infty} c'(x). \end{aligned}$$

The positive autarky effort k^* uniquely solves $b'(k^*) = c'(k^*)$.

The benchmark of Bramoullé and Kranton (2007) has $c(x) = x$ and $\delta = 1$.

SET, Section 4.2, p. 38, equation (48) and Assumption 2

Best responses, corners, and existence

Write $t_i = \delta \sum_j g_{ij} x_j$ and

$$\theta(t) = \arg \max_{z \geq 0} \{b(z + t) - c(z)\}.$$

Equilibrium conditions, including zero effort

$$0 \leq x_i \perp F_i(x) = c'(x_i) - b'(x_i + t_i) \geq 0.$$

Where the best response is positive,

$$\theta'(t) = \frac{b''(\theta(t) + t)}{c''(\theta(t)) - b''(\theta(t) + t)} \in [-1, 0).$$

Every best response lies in $[0, k^*]$. A continuous best-response map on $[0, k^*]^n$ therefore has a fixed point.

SET, pp. 38 and 50–51, equation (49) and proof of Theorem 5

A decreasing inverse produces a symmetric system

Let $q_i = x_i + \delta \sum_j g_{ij} x_j$ and, on the relevant inverse domain, define

$$h(x_i) = (b')^{-1}(c'(x_i)).$$

Since b' is strictly decreasing,

$$\operatorname{sgn}(c'(x_i) - b'(q_i)) = \operatorname{sgn}(q_i - h(x_i)).$$

The transformed operator

$$\tilde{F}_i(x) = x_i + \delta \sum_j g_{ij} x_j - h(x_i), \quad \frac{\partial \tilde{F}_i}{\partial x_j} = \delta g_{ij} \quad (j \neq i).$$

Rectangularity preserves complementarity. Symmetry of G gives integrability.

SET, p. 39, SET2 application preceding equation (50)

A potential for the transformed equilibrium conditions

Using the convex-minimization sign convention, define

$$\Phi(x) = \frac{1}{2}x^\top (I + \delta G)x - \sum_i \int^{x_i} h(t) dt.$$

$$\nabla \Phi(x) = \tilde{F}(x), \quad 0 \leq x \perp \nabla \Phi(x) \geq 0.$$

When the potential is convex

Equilibria solve $\min_{x \geq 0} \Phi(x)$.

Strict convexity gives at most one solution. Existence follows separately.

The integral and any boundary extension must be well defined on the domain used.

SET, p. 39, equation (50), with $\Phi = -\phi^{\text{BKC}}$

The smallest eigenvalue controls a uniqueness condition

The Hessian of the transformed potential is

$$\nabla^2 \Phi(x) = I + \delta G + D(x), \quad D_{ii}(x) = -\frac{c''(x_i)}{b''(h(x_i))} \geq 0.$$

Theorem 5: a sufficient spectral condition

Under Assumption 2 and the relevant transformed-domain conditions,

$$1 + \delta \lambda_{\min}(G) > 0 \implies \text{a unique Nash equilibrium.}$$

- Cost curvature adds $D \geq 0$, so this sufficient bound can be conservative.
- For linear costs, $D = 0$ and the bound is exact for strict convexity.

SET, Theorem 5 and Remark 12, pp. 39–40, proof on pp. 50–51

Zero marginal cost at zero implies full participation

Now suppose c is strictly convex, $c'(0) = 0$, and $\delta = 1$, as in Proposition 7.

Every equilibrium is interior

At any finite profile with $x_i = 0$,

$$\left. \frac{\partial u_i}{\partial x_i} \right|_{x_i=0} = b' \left(\sum_j g_{ij} x_j \right) - c'(0) > 0.$$

Hence $x_i^* > 0$ for every player at every Nash equilibrium.

All equilibrium conditions therefore hold with equality:

$$b' \left(x_i^* + \sum_j g_{ij} x_j^* \right) = c'(x_i^*).$$

Full participation does not require equilibrium uniqueness.

SET, Assumption 3 and Proposition 7(i), p. 40, proof on p. 51

Complete networks: convex costs remove the continuum

Take the complete network and $\delta = 1$. Every player receives total provision $S = \sum_j x_j$.

Strictly convex costs, $c'(0) = 0$

$$c'(x_i) = b'(S) \quad \text{for all } i.$$

Strictly increasing c' forces equal effort.

$$x_i = \bar{x}, \quad b'(n\bar{x}) = c'(\bar{x}).$$

The equilibrium is unique.

Linear costs, $c(x) = x$

With $b'(k^*) = 1$, every profile satisfying

$$x_i \geq 0, \quad \sum_i x_i = k^*$$

is an equilibrium.

The continuum includes both equal sharing and a single specialist.

Here $\lambda_{\min}(G) = -1$. Uniqueness with strictly convex costs holds even though $1 + \delta\lambda_{\min}(G) = 0$.

SET, Proposition 7(iii), p. 40, proof on p. 51

Network structure determines effort rankings

Maintain homogeneous b, c , strictly convex costs, $c'(0) = 0$, and $\delta = 1$. Let N_i and $\bar{N}_i = N_i \cup \{i\}$ denote open and closed neighborhoods.

Neighborhood inclusion

At every equilibrium,

$$N_i \subsetneq N_j \text{ or } \bar{N}_i \subsetneq \bar{N}_j \implies x_i^* > x_j^*.$$

- **Star**, $n \geq 3$. Leaves choose equal effort. The hub contributes less.
- **Nested split graph**. If $d_i < d_j$, then $x_i^* > x_j^*$.

Degree alone does not imply this ranking on arbitrary networks.

SET, Proposition 7(ii, iv) and Corollary 2, pp. 40–41, proofs pp. 51–52

What SET preserves, and what must still be checked

Question	Required argument
Equilibrium equivalence	Valid VI representation, full-domain sign equivalence, and appropriate domain geometry.
Existence	A compactness, coercivity, strong-P, or bounded-iteration argument.
Uniqueness	A suitable P property, strict monotonicity, or strict convexity.
Dynamics	A separate stability or convergence analysis for the proposed adjustment rule.

Synthesis of SET, Sections 2 and 5, pp. 8–17, 41–42

Main conclusions

A general method

On rectangular domains, the signs of the equilibrium operator determine the VI solution set. Suitable transformations can reveal structure absent from the original Jacobian.

- CES complements combine a SET uniqueness argument with a Perron existence threshold.
- Bonacich centrality is a resolvent response. Eigenvector centrality describes its critical direction.
- Under strategic substitutes, cost curvature and the smallest network eigenvalue govern uniqueness.

Which transformation exposes the useful structure in a given model?

Backup: a formal coordinate proof of Theorem 1

Let $K = \prod_i K_i$ and suppose $x^* \in \text{Sol}(K, F)$. For every i and $t \in K_i$,

$$(t - x_i^*)F_i(x^*) \geq 0.$$

Sign equivalence implies the same inequality for $\tilde{F}_i(x^*)$:

$$\begin{cases} F_i(x^*) = 0 & \implies \tilde{F}_i(x^*) = 0, \\ F_i(x^*) > 0 & \implies t - x_i^* \geq 0 \text{ and } \tilde{F}_i(x^*) > 0, \\ F_i(x^*) < 0 & \implies t - x_i^* \leq 0 \text{ and } \tilde{F}_i(x^*) < 0. \end{cases}$$

Consequently, for every $x \in K$,

$$\langle x - x^*, \tilde{F}(x^*) \rangle = \sum_i (x_i - x_i^*) \tilde{F}_i(x^*) \geq 0.$$

Thus $\text{Sol}(K, F) \subseteq \text{Sol}(K, \tilde{F})$. Interchanging the two operators gives the reverse inclusion.

Backup: matrix tests in the payoff-scaling example

For $J = \begin{pmatrix} 1 & -\delta \\ -\kappa\delta & \kappa \end{pmatrix}$ with $\kappa > 0$:

P-matrix test

The principal minors are

$$1, \quad \kappa, \quad \kappa(1 - \delta^2).$$

All are positive if $0 < \delta < 1$.

Strict monotonicity test

For $S = (J + J^\top)/2$,

$$\det S = \kappa - \frac{(1 + \kappa)^2 \delta^2}{4}.$$

$S \succ 0$ iff $\delta < 2\sqrt{\kappa}/(1 + \kappa)$.

Let $\Delta = (1 - \kappa)^2 + (1 + \kappa)^2 \delta^2$. The smallest eigenvalue is

$$\lambda_{\min}(S) = \frac{1 + \kappa - \sqrt{\Delta}}{2}.$$

Rescaling by $\text{diag}(1, 1/\kappa)$ restores the symmetric derivative with eigenvalues $1 - \delta$ and $1 + \delta$.

SET, pp. 15–16; explicit principal-minor and eigenvalue calculations

Backup: the nonlinear Perron limit

Let $y(\delta) = x(\delta)^\beta$ and normalize $q(\delta) = y(\delta)/\|y(\delta)\|_1$.

- If actions remained bounded as $\delta \uparrow \bar{\delta}$, continuity would give an equilibrium at $\bar{\delta}$.
- Connectedness propagates divergence: $x_i \geq \delta x_j$ whenever $g_{ij} = 1$.

Therefore $a_i/x_i \rightarrow 0$, and along any convergent subsequence,

$$\frac{(x - a)^\beta}{\|x^\beta\|_1} = \delta^\beta Gq(\delta) \implies q = \bar{\delta}^\beta Gq.$$

Uniqueness of the normalized limit

The only nonnegative unit- ℓ_1 Perron vector is $e/\|e\|_1$. Every subsequential limit is the same, so the full sequence converges.

An expanded proof of SET, Theorem 4(ii), pp. 48–49

Backup: algebraic solutions and positive equilibria

For two linked agents with baseline $a = \mathbf{1}$,

$$G = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \lambda_1 = 1, \quad \lambda_2 = -1.$$

Whenever $\delta \neq \pm 1$, the linear system has the solution

$$x = (I - \delta G)^{-1} \mathbf{1} = \frac{1}{1 - \delta} \mathbf{1}.$$

Parameter	Algebraic system	Economic equilibrium
$0 < \delta < 1$	Unique solution	Positive and finite
$\delta = 1$	Singular, inconsistent	None
$\delta > 1$	Unique negative solution	None on $\mathbb{R}_+^2$

Illustrative calculation for the linear special case of SET

Backup: complete-network uniqueness without the strict spectral bound

Assume homogeneous b, c , strictly convex c , $c'(0) = 0$, and $\delta = 1$. On the complete network, write $S = \sum_j x_j$.

1. Full participation makes every first-order condition an equality:

$$c'(x_i) = b'(S) = c'(x_j) \quad \text{for every } i, j.$$

2. Strictly increasing c' implies $x_i = x_j = \bar{x}$.
3. The function $f(z) = b'(nz) - c'(z)$ is strictly decreasing, with $f(0) > 0$ and an eventually negative value. Thus it has exactly one positive zero.

Why the spectral test is only sufficient

For $n \geq 2$, the complete graph has $\lambda_{\min}(G) = -1$. At $\delta = 1$ the strict spectral inequality fails, but the argument above proves uniqueness.

Backup: why the hub contributes less in a star

Let $n \geq 3$ and $\delta = 1$, with homogeneous functions, strictly convex costs, and $c'(0) = 0$. Write x_H for the hub's effort.

1. Every leaf faces the same peer effort x_H and has a unique best response. Thus all leaves choose one common effort $x_L > 0$.

2. The hub and a leaf satisfy

$$c'(x_H) = b'(x_H + (n-1)x_L), \quad c'(x_L) = b'(x_H + x_L).$$

3. Since $(n-2)x_L > 0$ and b' is strictly decreasing,

$$c'(x_H) < c'(x_L).$$

Strictly increasing c' then gives $x_H < x_L$.

The proof applies to every equilibrium, without a uniqueness assumption.

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The seminar follows the supplied SET draft. Slide citations refer to its printed page numbers. The path-network curves and two-node calculation are seminar examples.